

Section A

- 1 (a) (i) force proportional to product of masses B1
force inversely proportional to square of separation B1 [2]
- (ii) separation much greater than radius / diameter of Sun / planet B1 [1]
- (b) (i) e.g. force or field strength $\propto 1 / r^2$
potential $\propto 1 / r$ B1 [1]
- (ii) e.g. gravitational force (always) attractive B1
electric force attractive or repulsive B1 [2]

2 (a) number of atoms of carbon-12

M1